

STRUCTURE OF THE CUMBERLAND PLATEAU, TENNESSEE

BY CHARLES W. WILSON, JR., AND RICHARD G. STEARNS

ABSTRACT

Present structure of the Cumberland Plateau results from two components: (1) regional eastward dip of about 25 feet per mile off the Cincinnati arch; and (2) Pine Mountain, Cumberland Plateau, and Sequatchie Valley bedding thrusts, which strike north-eastward and have roots in the Valley and Ridge province to the southeast.

Gentle eastward dip off the flank of the Cincinnati arch is the only structure affecting the undisturbed portion of the plateau, and is also the primary component of structure in the faulted area. Bedding thrusts do not disturb the position and attitude of strata within the thrust sheets. Locally, however, where the faults cut across bedding, superficial anticlines are formed. The structural concordance between the undisturbed areas and areas of bedding thrusts supports the "no-basement" interpretation for the essential regional structure.

Movement of the Cumberland Plateau overthrust sheet over steep portions of its generally flat fault plane caused superficial flat-topped, steep-sided anticlines similar to the Powell Valley anticline of the Pine Mountain overthrust. Abrupt ends of such anticlines mark places where buried steep thrusts become buried strike-slip faults.

The oldest thrust is the northwest-dipping Whetstone Mountain fault; next is the Cumberland Plateau overthrust; the youngest thrusts are the Sequatchie Valley thrust and some faults of the Valley and Ridge province.

CONTENTS

TEXT	Page		Page
Introduction.....	1283	Summary and conclusions.....	1295
Previous investigations.....	1284	References cited.....	1296
Acknowledgments.....	1285		
Structure of the Cumberland Plateau.....	1286		
Pine Mountain overthrust.....	1286	ILLUSTRATIONS	
General statement.....	1286	Figure	Page
Pine Mountain block.....	1286	1. Regional structural features and location of	
Pine Mountain thrust fault.....	1286	the Cumberland Plateau.....	1284
Jacksboro cross fault.....	1287	2. Structural features of the Cumberland	
Cumberland Plateau overthrust.....	1290	Plateau, Tennessee.....	1285
General relationships.....	1290	3. Generalized Pennsylvanian stratigraphic	
Cumberland Plateau overthrust sheet.....	1290	column for the Cumberland Plateau,	
Outcropping faults of the overthrust system.....	1291	Tennessee.....	1287
Surficial anticlines caused by thrusting.....	1292	4. Structure contour map of the Cumberland	
Relationships of the Cumberland Plateau		Plateau, Tennessee.....	1288
overthrust to other structures.....	1293	5. Structure cross section of the Pine Moun-	
Sequatchie Valley anticline.....	1294	tain overthrust.....	1290
Eastern Cumberland escarpment.....	1294	6. Structure cross sections of the Cumberland	
Undisturbed area of the Cumberland Plateau	1294	Plateau overthrust and the Sequatchie	
		Valley anticline.....	1291
		7. Block diagrams of the Cumberland Plateau	
		overthrust.....	1292
		8. Mechanics of superficial anticline formation	1293

INTRODUCTION

The Cumberland Plateau extends across Tennessee from Kentucky and Virginia on the north to Georgia and Alabama on the south. Regional physiographic and structural features are shown on Figure 1. Although the domi-

nant structural pattern of the plateau is a gentle dip to the east off the Cincinnati arch, this slope is broken by the surface trace of the Pine Mountain and Cumberland Plateau overthrusts, the Sequatchie Valley anticline, and by an abrupt upturn along the eastern escarpment of the plateau (Fig. 2).

The Cumberland Plateau is underlain by Pennsylvanian rocks. Mississippian rocks crop out on lower slopes of the western escarpment, but beds as old as the Knox dolomite of Cambro-Ordovician age are exposed in Sequatchie

PREVIOUS INVESTIGATIONS

Safford (1869), the first to describe the Cumberland Plateau structure, recognized the Sequatchie Valley anticline, the Pine Mountain

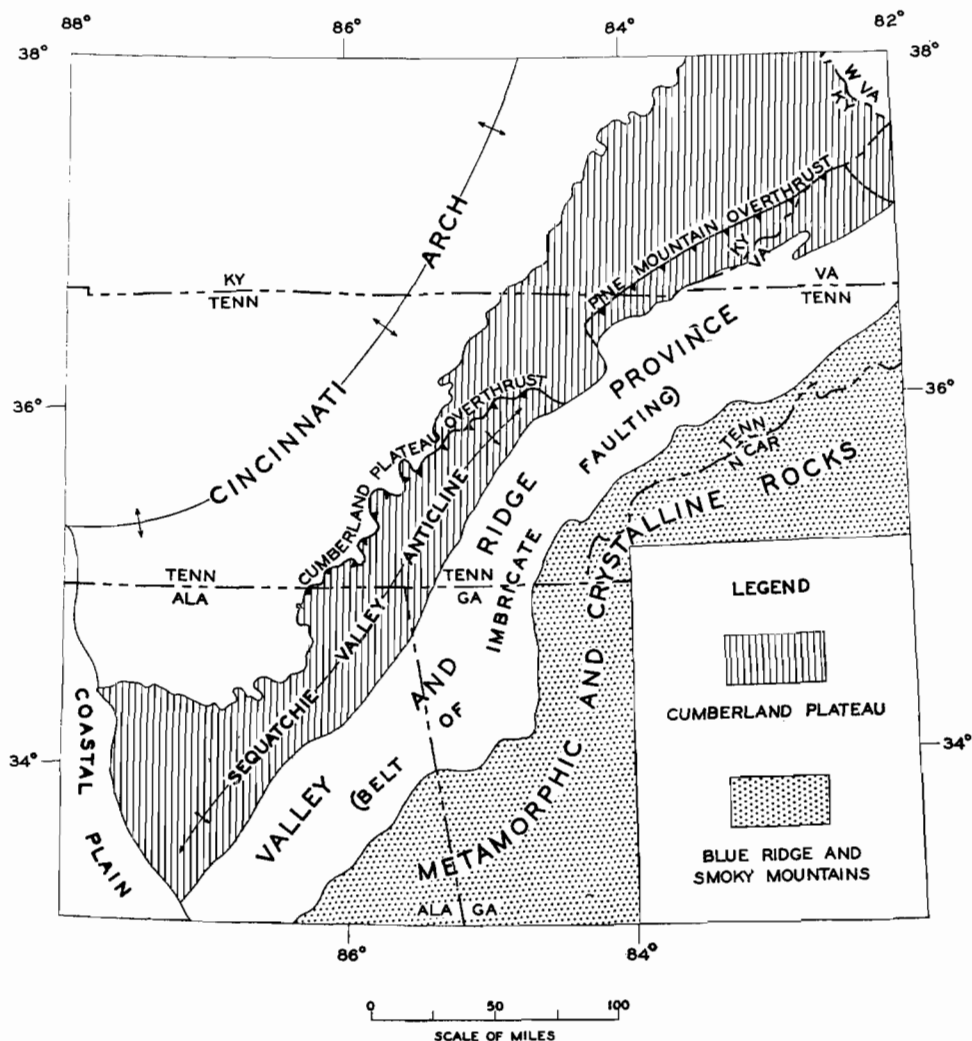


FIGURE 1.—REGIONAL STRUCTURAL FEATURES AND LOCATION OF THE CUMBERLAND PLATEAU

Valley, and Cambrian strata are brought up by thrusting along the eastern escarpment.

This report is primarily concerned with the configuration of Pennsylvanian rocks. Figure 3 shows the generalized stratigraphic column of the Pennsylvanian. Discussion of the structure which follows is arranged in geographic order by the structural subprovinces shown on Figure 2.

block, and the sharp upturn at the eastern edge of the plateau which marks its boundary with the Valley and Ridge province. He also noted disturbances now known to be along the trace of the Cumberland Plateau overthrust but considered them to be of only local significance.

More detailed descriptions include those of Hayes (1894a; 1894b; 1894c; 1895a; 1895b),

Keith (1896; 1897), Campbell (1899), Butts (1916; 1919; 1927), Butts and Nelson (1925), Glenn (1925), and Nelson (1925a; 1925b).

Wentworth (1921), Butts (1927), and Rich (1934) established that the Pine Mountain

near the Tennessee line. Stearns (1954; 1955) demonstrated that most of the southern Cumberland Plateau has moved northwestward along the relatively flat Cumberland Plateau overthrust in a manner similar to the Pine

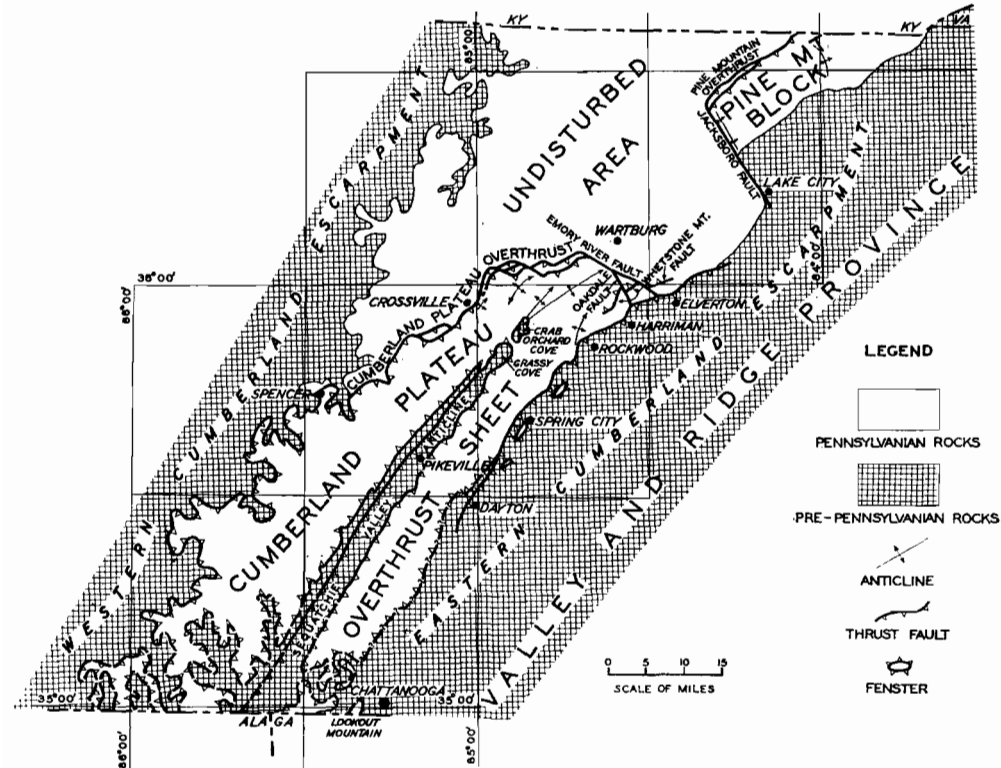


FIGURE 2.—STRUCTURAL FEATURES OF THE CUMBERLAND PLATEAU, TENNESSEE

block is part of a low-angle thrust sheet which has its roots in the Valley and Ridge province to the southeast. Rich (1934) suggested that the Sequatchie Valley anticline similarly resulted from low-angle thrusting. Rodgers (1950) presented new information supporting this concept and suggested that outlying structures northwest of the Sequatchie Valley anticline are also part of an incipient thrust sheet. His geologic map of East Tennessee (1953a) includes the southeastern boundary of the plateau. Rodgers (1953b) prepared a cross section of the combined Valley and Ridge and Cumberland Plateau which interprets the visible structures as confined to the near-surface rocks and not involving the lower strata or the basement. The present investigation mainly bears out his interpretation.

Miller and Fuller (1954) did detailed work on the Pine Mountain overthrust in Virginia,

Mountain overthrust to the north. Swingle *et al.* (1956) reported that Pennsylvanian beds, previously thought to be restricted to the plateau, lie buried by thrusting in adjacent areas of the Valley and Ridge province. They concluded that the Cumberland Plateau and Valley and Ridge province are not sharply divided but merge structurally. Wilson *et al.* (1956) presented structure maps and a summary of the structural features of the plateau; this information is part of a folio on general Pennsylvanian geology resulting from field work by many members of the Tennessee Division of Geology, including the present writers.

ACKNOWLEDGMENTS

The writers are indebted for structural information to many members of the Tennessee Division of Geology who participated in field

mapping of the Cumberland Plateau. Chief among these are John W. Jewell, Edward T. Luther, Robert M. Mitchum, Jr., and Robert L. Wilson.

Special thanks are due to W. D. Hardeman, State Geologist of Tennessee, who has maintained active support and interest in the geology of this area. He spent much time in the field and contributed many ideas and helpful suggestions.

The writers are grateful to George D. Swingle of the University of Tennessee and Herbert A. Tiedemann of the Tennessee Division of Geology for information on the eastern Cumberland escarpment and adjacent portions of the Valley and Ridge province.

Robert J. Floyd, Senior Geologist of the Tennessee Division of Geology, edited the manuscript.

STRUCTURE OF THE CUMBERLAND PLATEAU

Structurally, the Cumberland Plateau in Tennessee is divided into three subprovinces: (1) an area undisturbed by thrust faulting, comprising less than half the province; (2) the Pine Mountain overthrust sheet; and (3) the Cumberland Plateau overthrust sheet (see Fig. 2). Both the Pine Mountain and Cumberland Plateau overthrusts are thin sheets that moved primarily along extensive bedding thrusts. The thin Cumberland Plateau overthrust sheet is interrupted by the deeper-seated prominent Sequatchie Valley anticline. The southeastern boundary of the plateau is a prominent escarpment that locally is dominated by Valley and Ridge structure.

Figure 4 is a structural contour map of the Cumberland Plateau drawn on the Rockcastle conglomerate as a datum. This formation has been removed by erosion over much of the southern part of the plateau, where the configuration of the older Sewanee conglomerate has been projected to the Rockcastle.

PINE MOUNTAIN OVERTHRUST

General Statement

The Pine Mountain overthrust bounds a rectangular thrust sheet extending 125 miles from northeast Tennessee into Kentucky and Virginia. The northwest portion of the overthrust sheet is a broad syncline, underlain by Pennsylvanian rocks (herein called the Pine Mountain block), that is part of the Cumberland Plateau; the southeast part is the broad,

flat-topped Powell Valley anticline in the Valley and Ridge province (Fig. 1); Cambrian and Ordovician rocks are exposed at the surface of the anticline. The southwestern end of the thrust sheet moved about 10 miles. Displacement diminishes northeastward; near the Tennessee border in Virginia, the movement was only 5.8 miles (Miller and Fuller, 1954). Butts (1927), Rich (1934), and Miller and Fuller (1954) have discussed the structure and mechanics of this block in detail. Figure 5 shows a cross section of the Pine Mountain overthrust near its southwest end. The location of this cross section is plotted on Figure 4.

Pine Mountain Block

The Pine Mountain block refers only to that part of the larger thrust sheet that involves Pennsylvanian rocks. It is separated from the main body of the plateau by the Pine Mountain and Jacksboro faults. The Pine Mountain block is surrounded by prominent strike ridges. It is bounded on the southeast by a steep northwest dip along the eastern Cumberland escarpment. This is the steep northwest flank of the Powell Valley anticline. On the northwest is Pine Mountain, which reflects a 10- to 35-degree southeast dip of resistant beds on the hanging wall of the Pine Mountain fault. To the southwest, on the northeast side of the Jacksboro fault, is a prominent anticlinal ridge. Along all the strike ridges at the periphery of the block, dip flattens abruptly to merge with the relatively flat floor of the syncline.

Within the block is a rectangular structural basin with nearly flat beds throughout most of its extent. It is, however, underlain by a bedding thrust, which has its roots in the Valley and Ridge province to the southeast. Along this thrust Pennsylvanian strata have been moved northwestward about 10 miles. Corresponding Pennsylvanian beds within the block are about 500 feet above their normal level on the adjacent parts of the plateau.

Pine Mountain Thrust Fault

At the trace of the Pine Mountain fault, beds as old as Silurian have been thrust over Pennsylvanian. The fault extends more than 100 miles beyond Tennessee into Kentucky and Virginia. Northwest of the Pine Mountain thrust in Tennessee is a subsidiary fault slice which consists of intensely deformed beds of the Gizzard and Crab Orchard Mountains

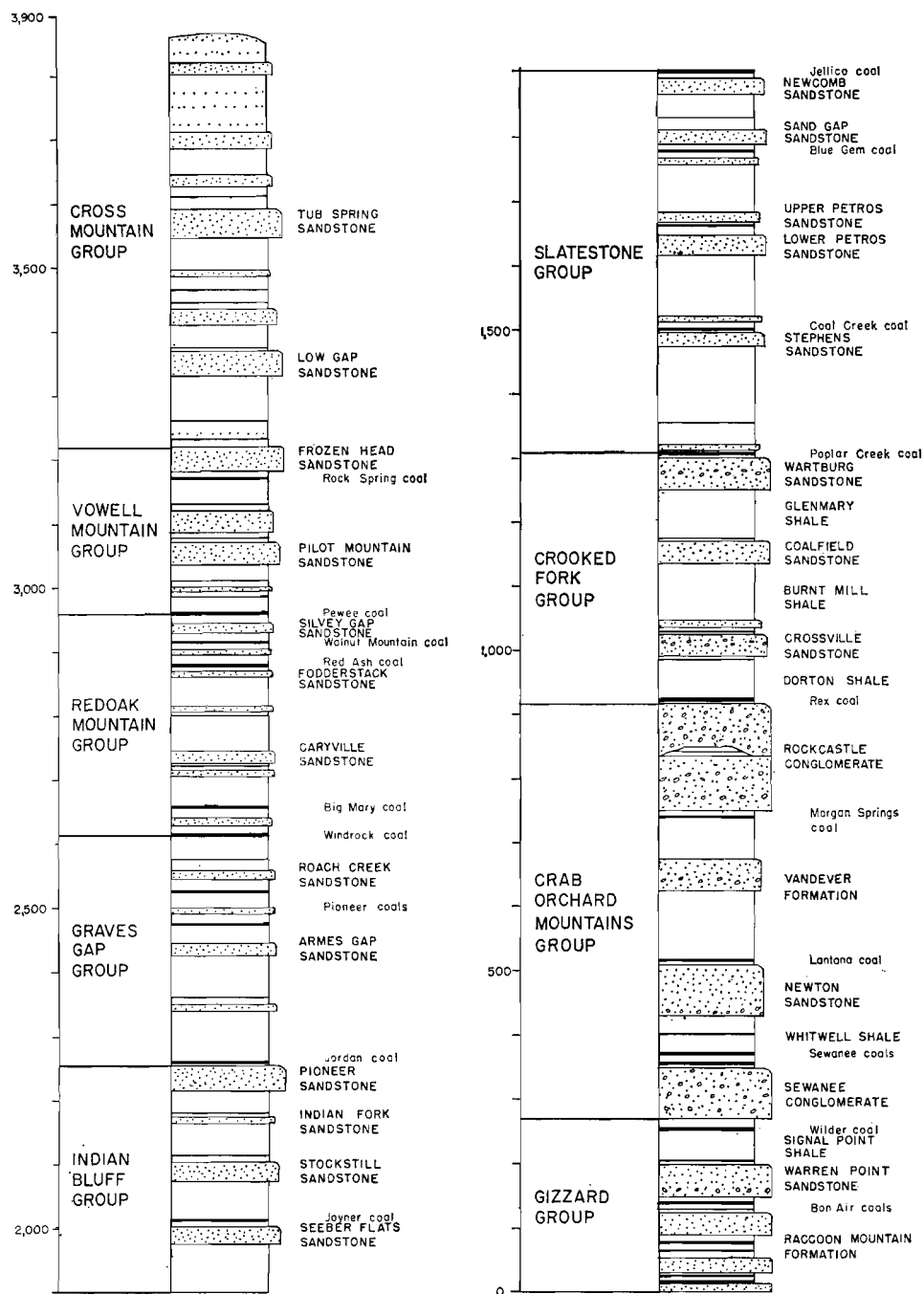


FIGURE 3.—GENERALIZED PENNSYLVANIAN STRATIGRAPHIC COLUMN FOR THE CUMBERLAND PLATEAU, TENNESSEE

groups. The gentle dip of this fault is shown by an outlying klippe known as Hells Point Ridge, where brecciated older sandstones occur above flat-lying younger coal-bearing beds (see Fig. 5).

Jacksboro Cross Fault

The southwestern boundary of the Pine Mountain block is the Jacksboro cross fault.

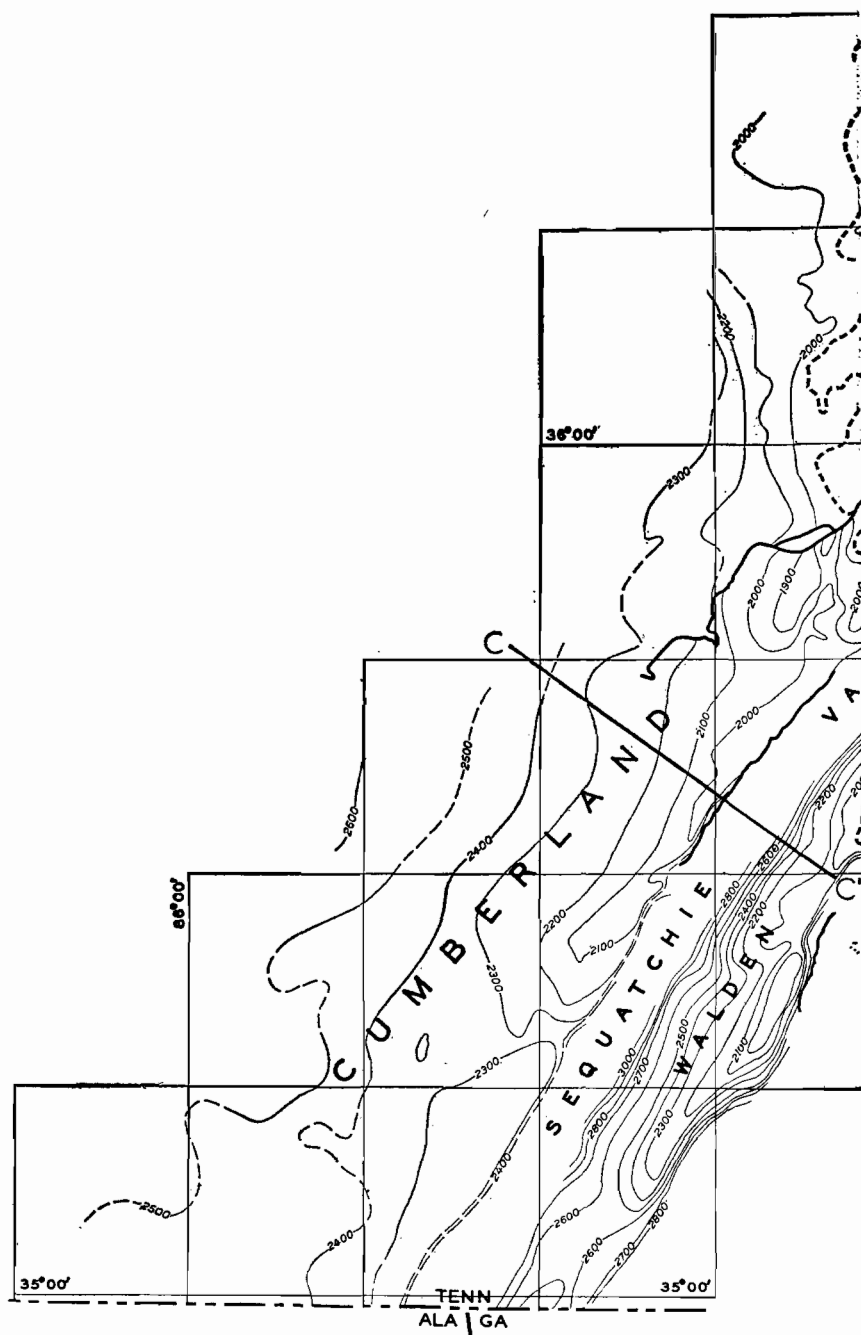
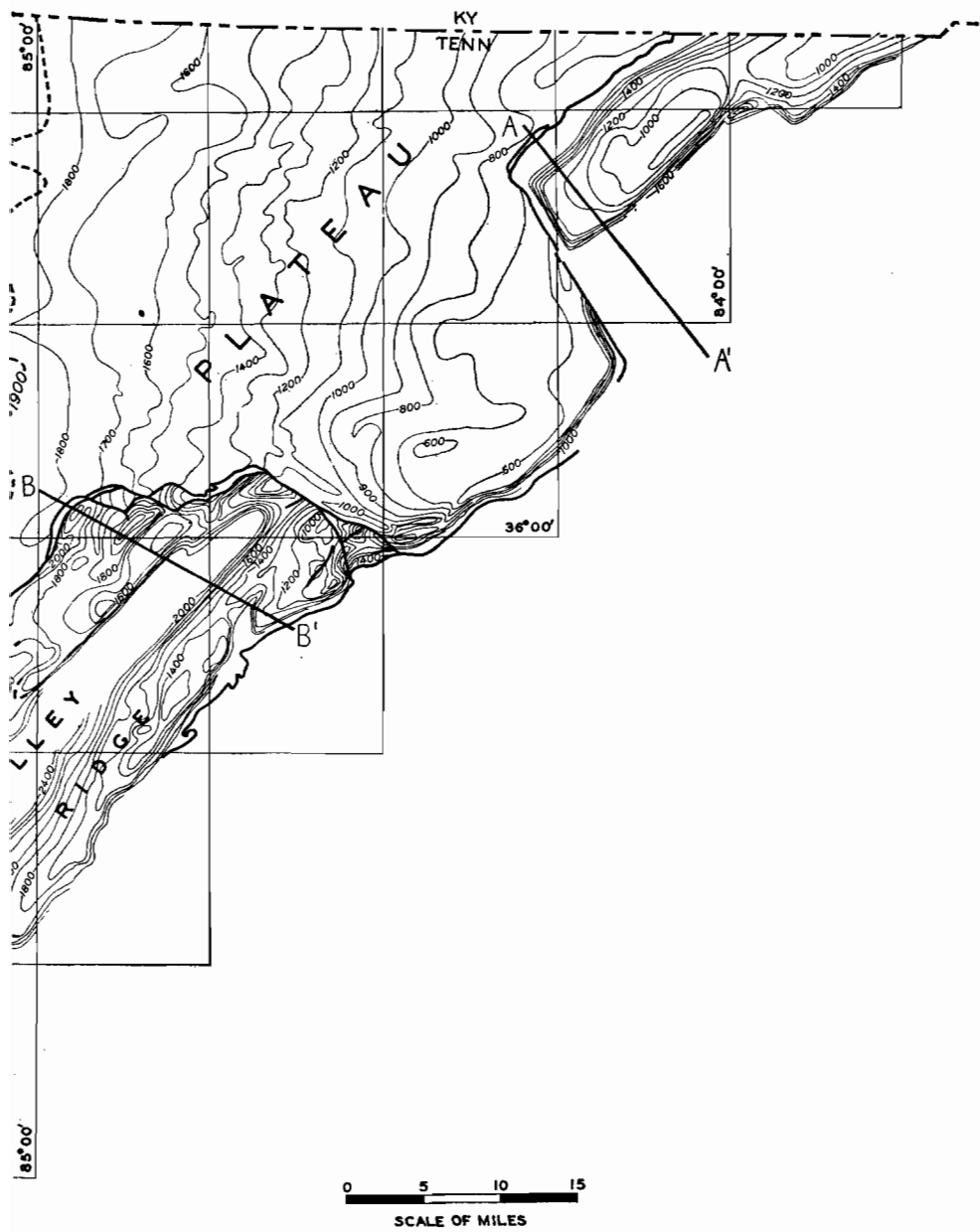


FIGURE 4.—STRUCTURE CONTOUR MAP OF
Lines A-A', B-B', and C-C' locate struc-



THE CUMBERLAND PLATEAU, TENNESSEE
ure cross sections, Figures 5 and 6.

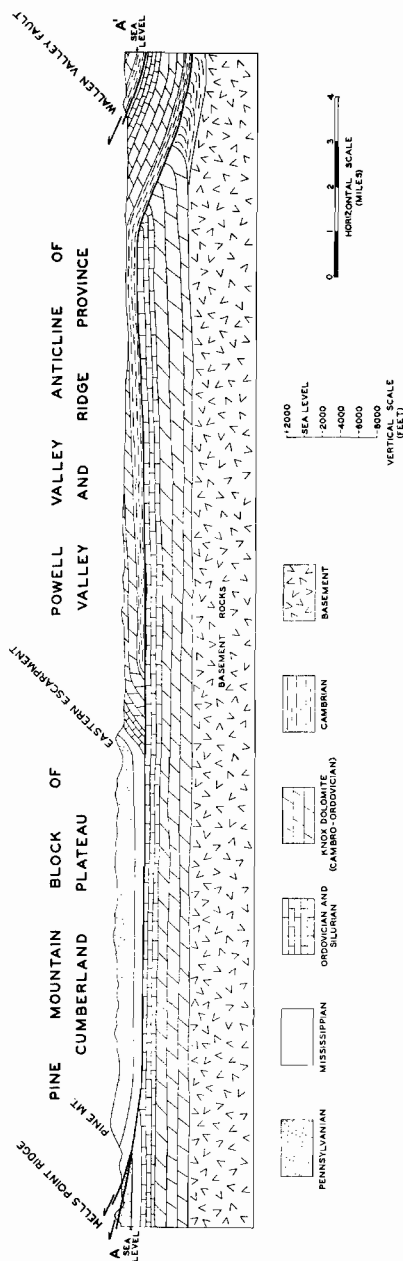


FIGURE 5.—STRUCTURE CROSS SECTION OF THE PINE MOUNTAIN OVERTHRUST Section is along line A-A' of Figure 4. Modified after Rich (1934), and Rodgers (1953a). Horizontal and vertical scales are the same.

This fault is nearly vertical at the surface, but northeastward, in the subsurface, it flattens abruptly and merges with the Pine Mountain thrust. This fault moved laterally approxi-

mately 10 miles, but its vertical movement was only about 500 feet. The "crowding" accompanying lateral movement resulted in the tight anticline along the northeast side of the fault (Fig. 2). Here, the deformed beds of the Crab Orchard Mountains and Gizzard groups are faulted against nearly flat Slatestone beds (Fig. 3). Southeast of the Pine Mountain block, the Jacksboro fault extends along the plateau edge as far as Lake City (Fig. 2).

CUMBERLAND PLATEAU OVERTHRUST

General Relationships

The Cumberland Plateau overthrust limits a thrust sheet, similar to the Pine Mountain block, which includes the entire southern half of the plateau. The northern limit of the overthrust sheet is a complex series of faults, which crosses the plateau from Elverton on the east to Spencer on the west (Fig. 2). The sheet continues southwestward as a bedding thrust along the western escarpment face at least as far as the southern border of Tennessee. It also crops out as a bedding fault around Sequatchie Valley, Grassy Cove, and Crab Orchard Cove, making fensters of these valleys.

Regionally, movement of the Cumberland Plateau overthrust sheet appears to have been rotational, diminishing southwestward, mirroring the rotational movement of the Pine Mountain overthrust sheet which diminishes northeastward (Butts, 1927). This thrust sheet is known to extend nearly to the Alabama border; the writers believe that it continues southward into Alabama, although the movement may have been slight.

Cumberland Plateau Overthrust Sheet

Like the Pine Mountain block, this southern part of the plateau is in general underlain by gently dipping strata (Fig. 4). Also, like the Pine Mountain block, this extensive area of nearly flat beds is completely underlain by bedding thrusts of the Cumberland Plateau overthrust. Pennsylvanian beds have been detached from underlying rocks and thrust as far as 3 miles northwestward. A major interruption of this area is the Sequatchie Valley anticline, which strikes northeast throughout the entire length of the thrust sheet.

Between the Sequatchie Valley anticline and the eastern escarpment is an asymmetrical syncline, with the axis close to the eastern escarpment, except in the southernmost part

near Chattanooga, where it is more symmetrical. This syncline includes a series of elongate gentle basins; it generally pitches northeastward, and the lowest structural position within the Cumberland Plateau overthrust sheet occurs in this syncline east of the end of the Sequatchie Valley anticline. Here,

Emory River fault that intersects the eastern escarpment at Harriman. This fault is also overridden by a later thrust of the Valley and Ridge province (G. D. Swingle, personal communication). Other individual thrust and cross faults of this system have been named and described in detail by Stearns (1954).

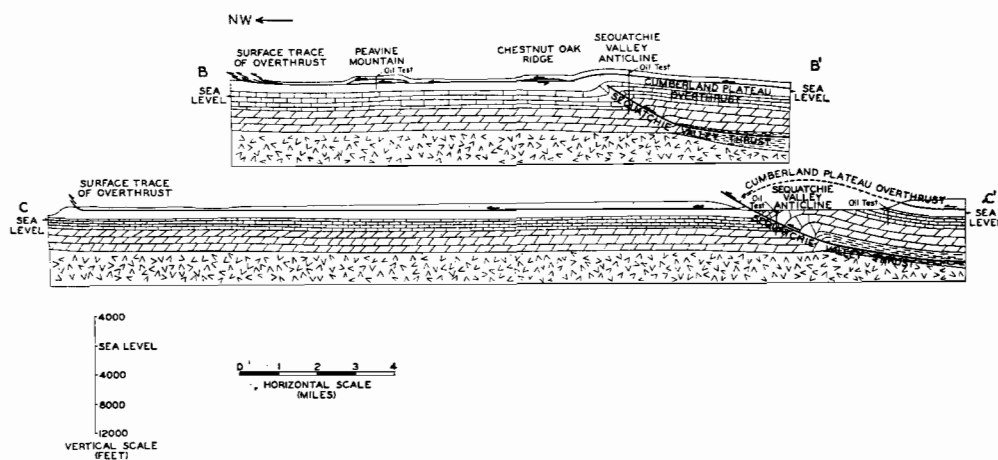


FIGURE 6.—STRUCTURE CROSS SECTIONS OF THE CUMBERLAND PLATEAU OVERTHRUST AND THE SEQUATCHIE VALLEY ANTICLINE
Sections along lines B-B' and C-C' of Figure 4. Horizontal and vertical scales are the same.

the syncline axis is about 1400 feet lower than it is at the Georgia-Tennessee border. Similarly, a syncline, the axis of which lies just west of the Sequatchie Valley anticline, pitches northeastward from the Alabama-Tennessee border to the Emory River fault where the strata are about 1300 feet lower. (See Fig. 4.)

Outcropping Faults of the Overthrust System

From Elverton to Spencer, across the plateau, the overthrust system crops out as a series of echelon thrusts joined by vertical cross faults. At the surface these thrusts dip more than 30° southeast, but the cross faults are vertical. Along the western escarpment south of Spencer, and around Sequatchie Valley, Grassy Cove, and Crab Orchard Cove, the system crops out as bedding thrusts.

The Emory River fault, first described by Jillson (1923), is the northeast limit of the Cumberland Plateau overthrust. It is the counterpart of the Jacksboro fault of the Pine Mountain overthrust. This fault moved laterally about a mile. The Emory River fault is overridden by Valley and Ridge thrusts at the escarpment. The Oakdale fault is a branch of the

Thrusts and cross faults are portions of the same complex fault plane, because they do not offset one another, but turn and merge. Thrusts in this belt are easily traced owing to the prominent ridges formed from tilted resistant sandstone formations, but cross faults are traced with difficulty because of lack of stratigraphic offset. Southwest of Spencer and around Sequatchie Valley, where the fault is a bedding thrust, the only effect is contortion and shearing of shales and coal in zones a few feet thick.

The Cumberland Plateau overthrust extends through a sequence of Pennsylvanian sandstones and conglomerates interlayered with incompetent coal-bearing shales. As a bedding thrust it follows the incompetent shales over extensive areas (Fig. 6); as a thrust or a cross fault it breaks through competent sandstones along narrow zones.

The configuration of the fault plane is known from exposures in fensters (Sequatchie Valley, Grassy Cove, and Crab Orchard Cove) and along the fault trace to the north and west. In these exposures the fault is a bedding thrust in various incompetent shaly zones. Its position rises stratigraphically to the northwest in the direction of fault movement, and

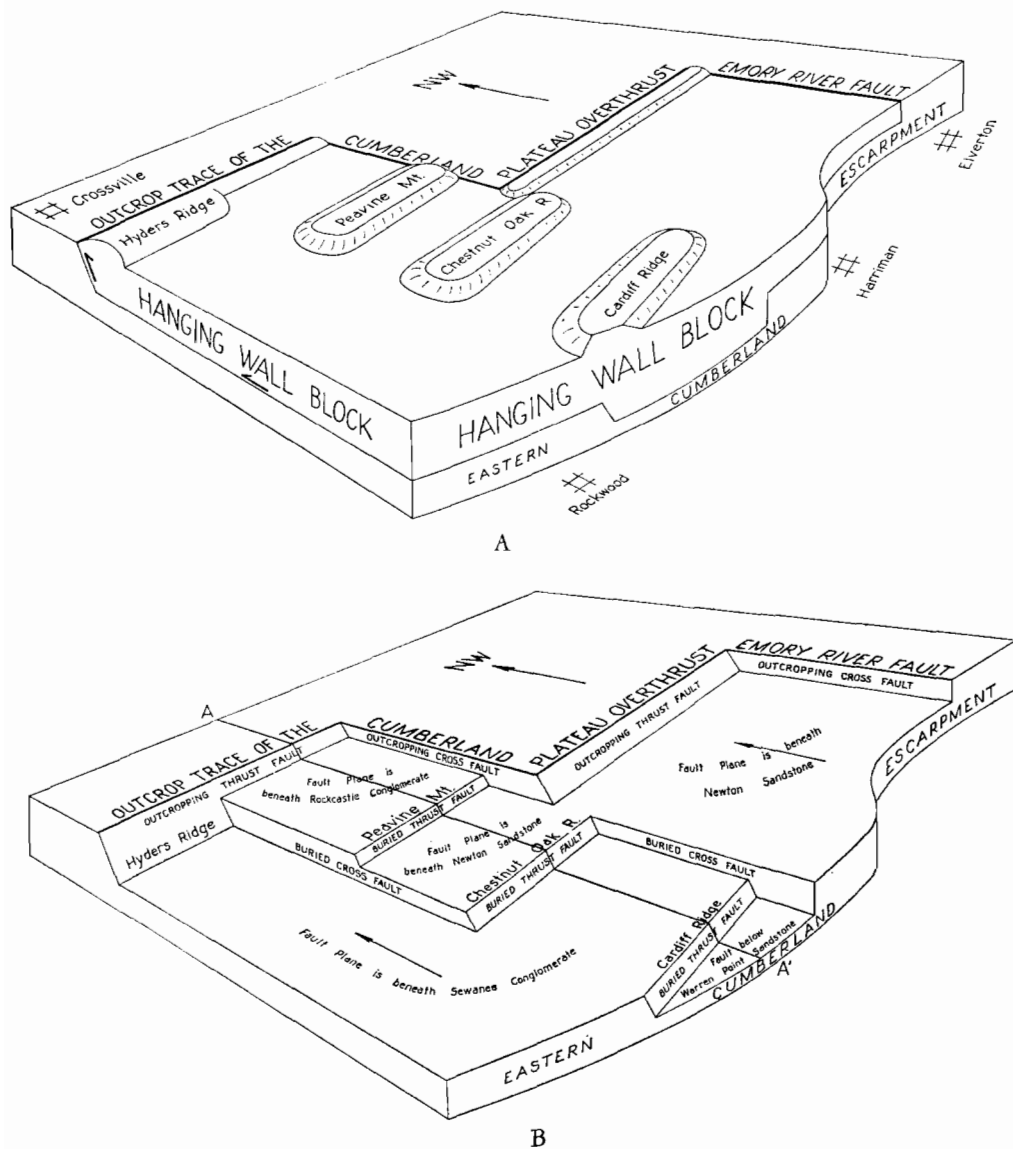


FIGURE 7.—BLOCK DIAGRAMS OF THE CUMBERLAND PLATEAU OVERTHRUST
 A shows the location of the surficial anticlines. B shows the configuration of the fault plane. The thrust sheet, or hanging wall block, is removed in B, exposing the footwall.

to the northeast across the direction of movement.

Surficial Anticlines Caused By Thrusting

Drilling has revealed that Peavine Mountain, surficially an anticline, is underlain at a shallow depth by flat beds. Uplift is due to subsurface duplication of beds where the Cumberland Plateau overthrust breaks through the competent Newton sandstone. The main fault

crosses out 5 miles northwest of Peavine Mountain, where it thrusts Rockcastle conglomerate over Crossville sandstone. Peavine Mountain has a broad, flat top and steep sides similar to the Powell Valley anticline of the Pine Mountain overthrust (Fig. 5). This square shape in cross section is characteristic of anticlines formed over a steep segment of a buried low-angle thrust, along which there has been considerable movement (Rich, 1934).

The subsurface configuration of the Cumber-

land Plateau overthrust can be inferred from the fault exposures and from the position of Peavine Mountain and similar anticlines. Buried steep thrusts are marked by the anticlines, but buried cross faults leave no surface

as shown on Figure 6. Had there been any arching of this anticline before the overthrusting occurred, the overthrust would have broken through to the surface along the Sequatchie Valley anticline.

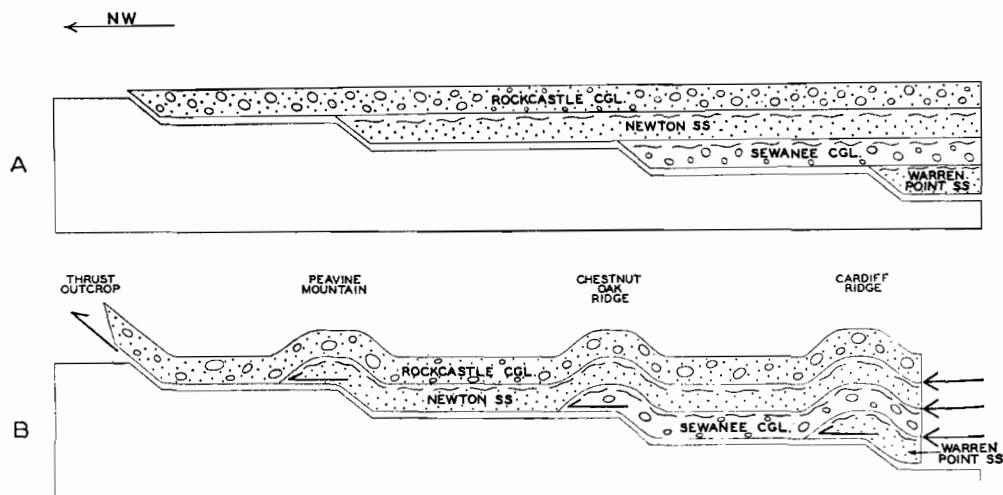


FIGURE 8.—MECHANICS OF SUPERFICIAL ANTICLINE FORMATION

A is before thrusting, B is after thrusting. The location of this cross section is shown on Figure 7B (section A-A').

indication. The 3 miles of movement responsible for Peavine Mountain structure cannot end abruptly, as does the anticline itself, without a cross fault. The end of the anticline, which marks the end of a buried thrust, also marks the junction of this buried thrust with a buried cross fault.

Figure 7A is a block diagram showing the relative positions of the thrust outcrop and the surficial anticlines. On Figure 7B the thrust plate is removed to show the buried fault-plane configuration of the Cumberland Plateau overthrust interpreted from the position of the anticlines and from exposures of the fault. Figure 8 is a diagrammatic cross section showing the mechanics of formation of three anticlines. Structure of the eastern escarpment and the northeast end of the Sequatchie Valley anticline is omitted on Figures 7 and 8 to simplify the relationships of the thrust plate.

Relationships of the Cumberland Plateau Overthrust to Other Structures

The Cumberland Plateau overthrust predates the Sequatchie Valley anticline, because it is folded by the Sequatchie Valley anticline and is offset by the Sequatchie Valley thrust,

Prior to movement of the Cumberland Plateau overthrust, the Whetstone Mountain fault was formed parallel to Valley and Ridge structure (Fig. 2). It is exceptional because the fault plane dips northwest. This structure has been offset approximately 1 mile by the Emory River fault of the Cumberland Plateau overthrust and has probably been offset a lesser distance by the Oakdale fault (Fig. 2).

The northwest-dipping Whetstone Mountain fault is the oldest structure described; the Cumberland Plateau overthrust is the next oldest; and the Sequatchie Valley anticline and at least some of the Valley and Ridge thrusts are the youngest.

Another low-angle overthrust, named the Ozone thrust (Stearns, 1954), occurs within the Cumberland Plateau overthrust mass and crops out east of Crab Orchard Cove, where the Newton sandstone is repeated. The fault is not shown on the illustrations, because its extent is unknown. The Ozone fault is the same type of structure as the main overthrust, and future work may reveal other bedding thrusts. This fault probably is approximately the same age as the Cumberland Plateau overthrust.

SEQUATCHIE VALLEY ANTICLINE

This long, straight anticline extends 200 miles from Blount County, Alabama, northeastward into Morgan County, Tennessee (Fig. 1). The southeastern flank dips gently, but the northwestern flank is steep and broken by the Sequatchie Valley thrust throughout most of the length of Sequatchie Valley. Along this thrust, beds as old as Cambro-Ordovician are carried over Pennsylvanian strata. The anticline pitches northeastward and terminates at the Emory River fault (Fig. 2).

From the Alabama-Tennessee state line to a point 10 miles north of Pikeville, resistant Pennsylvanian strata along the crest of this anticline have been removed by erosion, exposing less-resistant carbonate rocks into which the prominent Sequatchie Valley has been cut (Fig. 2). Northeast of the head of Sequatchie Valley, Pennsylvanian strata have been breached locally, and erosion has cut into the underlying Mississippian beds to form two valleys, Grassy Cove to the south and Crab Orchard Cove to the north (Fig. 2). Still farther northeast, the Crab Orchard Mountains mark the unbreached portion of the diminishing Sequatchie Valley anticline.

The Sequatchie Valley fault, as a surface feature, ends near the head of the main valley. The Shell Oil Company's Peterson No. 1 well adjacent to Crab Orchard Cove has revealed that 14 miles beyond the northeast end of the surface fault the anticline is faulted at depth (B-B', Fig. 6). In this well, about 1750 feet of Middle Ordovician beds are repeated. This amount of repetition is significant, because it accounts for the entire surface structural relief of the anticline in this area.

The Cumberland Plateau overthrust is folded by the Sequatchie Valley anticline (as shown in sections B-B' and C-C', Fig. 6) and is offset by the Sequatchie Valley fault (section C-C', Fig. 6).

The isolated position, straightness, and length of this anticline have long been regarded as proof that basement rocks are involved in its structure at depth. Rich (1934), however, proposed that the anticline results from low-angle thrusting through Cambrian shales with roots far to the east in the Valley and Ridge province, and that the underlying basement is undisturbed. Rodgers (1950) emphasized this and pointed to cross faults at either end of the anticline as proof. However, the Emory River cross fault at the northeast end of the anticline is now known to be part of the superficial older

Cumberland Plateau overthrust rather than part of the Sequatchie Valley structure. The present writers agree with Rodgers and Rich and believe that the Sequatchie Valley fault is a low-angle overthrust in the subsurface like the Pine Mountain and Cumberland Plateau overthrusts, as shown on Figure 6.

Other writers have postulated that some of the low-angle thrusts of the Valley and Ridge province occur as bedding faults in Rome and Conasauga shales. During the Middle Ordovician the future site of Sequatchie Valley occupied a foreland shelf position and was not in the Appalachian geosyncline (Wilson, 1949, p. 323). It is believed likely that the Rome and lower Conasauga shales, which were restricted to the geosyncline, overlapped against the foreland near the present position of Sequatchie Valley. This could account for the localization of the Sequatchie Valley structure, the bedding fault being deflected upward where the zone of easy slippage ends. (See Fig. 6.)

EASTERN CUMBERLAND ESCARPMENT

A prominent ridge marks the entire southeastern limit of the Cumberland Plateau. The eastern escarpment on the Pine Mountain overthrust is an expression of steeply dipping Pennsylvanian beds forming the northwest limb of the Powell Valley anticline (Fig. 5). The eastern escarpment southeast of the undisturbed area (Fig. 2) also turns up sharply, but here Valley and Ridge faulting lies at the base of the escarpment. The structures here are probably due to crowding against Valley and Ridge faults, where the unyielding part of the plateau abuts the valley.

Southwest of Elverton, the escarpment includes the eastern continuation of the Cumberland Plateau overthrust. Here, the eastern escarpment is locally an anticline. The southeast limb of this feature is commonly buried or faulted out by Valley and Ridge thrusts. Fences between Dayton and Rockwood (Fig. 2) show that Pennsylvanian beds extend farther southeastward beneath Valley and Ridge thrusts.

UNDISTURBED AREA OF THE CUMBERLAND PLATEAU

West of the Pine Mountain block and north of the Cumberland Plateau overthrust the plateau is undisturbed by bedding thrusts (Fig. 2). Between the two thrust sheets the undisturbed area extends to the edge of the Valley

and Ridge province, and beds are upturned abruptly only at the eastern escarpment. Although this block is separated from the rest of the plateau by faulting at the surface, it is continuous at depth with beds beneath the bedding thrusts of the Cumberland Plateau overthrust and the Pine Mountain block. Also, it is believed to be continuous at depth with the Pennsylvanian fensters between Dayton and Rockwood in adjacent areas of the Valley and Ridge province.

On Figure 4 contours within the undisturbed area strike north, discordant to the northeast strike of the overthrust features and Valley and Ridge structure. This pattern also strongly influences the structure of the disturbed area in the southern Cumberland Plateau, because it continues beneath the thrusts. The southern Cumberland Plateau has a complex configuration owing to local uplift by thrusting, but between the uplifted areas thrusting is by bedding slippage only. These areas of bedding slippage occupy the same structural position they would have if there had been no bedding thrusts.

For example, on Figure 4 the 1900-foot contour line (dashed line) trends northward near the west edge of the plateau in the undisturbed area. South of the undisturbed area, it crosses synclines in places which line up as a continuation of the north trend in the undisturbed area. Other contours may be similarly projected. Also, in the valley, beds in the fensters are at approximately the same elevation as they would be without bedding thrusts. This concordance between the structure of the undisturbed and disturbed areas of the Cumberland Plateau and adjacent portions of the valley is additional verification of the interpretation of the structure of the region that considers that basement rocks are not disturbed.

The gentle east dip off the flank of the Cincinnati arch, which is the only structure affecting the undisturbed area, is also the most important component of structure in the faulted area. Superficial upon this pattern in the southern Cumberland Plateau and Pine Mountain block are the anticlines and fractures due to bedding thrusts.

SUMMARY AND CONCLUSIONS

Present configuration of Cumberland Plateau strata results from two major structural components—the eastward regional dip off the flank of the Cincinnati arch, and bedding thrusts that have roots in the Valley and

Ridge province southeast of the plateau. These outlying bedding thrusts are the Pine Mountain overthrust, Cumberland Plateau overthrust, and Sequatchie Valley thrust. The area they underlie includes more than half the plateau. Even though thrusting is extensive, the whole plateau is underlain by nearly flat beds, which in most places dip about 25 feet per mile eastward off the Cincinnati arch. The only surface indications of the thrusts are dip and contortion in narrow zones adjacent to the fault trace, or anticlines such as Peavine Mountain, Chestnut Oak Ridge, and Cardiff Ridge.

The rectangular Pine Mountain block of northeast Tennessee, Kentucky, and Virginia is part of an overthrust sheet moved 10 miles northwestward on a bedding thrust in the Chattanooga shale and underlying Silurian beds. To the south the similar Cumberland Plateau overthrust sheet includes the entire width of the plateau. This thrust sheet is much thinner and near its northeast end has a maximum of 3 miles of displacement within the lower beds of the Pennsylvanian. Unlike the Pine Mountain overthrust sheet, it is not a rectangular thrust block. Its northern trace is an echelon system of thrusts and cross faults. It crops out as a bedding thrust on the face of the western escarpment. Sequatchie Valley, Grassy Cove, and Crab Orchard Cove are fensters. The Cumberland Plateau overthrust sheet extends to the Alabama border, where its amount of movement may be slight.

Movement of the Cumberland Plateau overthrust sheet over a steep portion of the fault plane is responsible for superficial anticlines, such as Peavine Mountain. These are flat-topped and steep-sided like the Powell Valley anticline of the Pine Mountain overthrust. They mark steep portions of the buried thrust where the fault breaks through competent formations in the subsurface, and the abrupt ends of such anticlines mark loci where buried thrusts turn and become buried strike-slip faults.

The Sequatchie Valley anticline is believed to be the result of a mile or so of low-angle thrusting, perhaps localized in its present position, because Cambrian shales pinch out there. The Sequatchie Valley anticline is faulted at the surface throughout most of its length, and drilling has revealed that the fault continues in the subsurface at least 14 miles beyond the end of the surface outcrop and probably to the end of the anticline.

The oldest structure is the northwest-dipping Whetstone Mountain thrust, which is offset by the northeastern cross faults of the Cumberland Plateau overthrust. Next in age is the Cumberland Plateau overthrust; this pre-dates the Sequatchie Valley anticline, because it is folded along with this anticline and offset by the Sequatchie Valley thrust. The Cumberland Plateau overthrust also pre-dates some of the faults of the Valley and Ridge province, because Valley and Ridge faults override the northeastern cross faults of this system.

Over most of the area underlain by bedding thrusts, the structure is as it would be if there were no thrusts. This concordance between the structure of the undisturbed and disturbed portions of the Cumberland Plateau is additional verification of the structural interpretation that considers the basement to be undisturbed. The easterly dip of 25 feet per mile off the flank of the Cincinnati arch is the only structural component affecting the undisturbed area; it is also the primary component of structure in the faulted area. Superficial upon this basic pattern in the southern Cumberland Plateau and Pine Mountain block are the complications of anticlines and fractures due to bedding thrusts.

REFERENCES CITED

- Butts, Charles, 1916, Structure of the southern part of Cumberland County, Tennessee, in relation to the possible occurrence of oil and gas: *Tenn. Geol. Survey, Resources of Tennessee*, v. 6, p. 107-110
- 1919, Geology and oil possibilities of the northern part of Overton County, Tennessee, and of adjoining parts of Clay, Pickett, and Fentress counties: *Tenn. Geol. Survey Bull.* 24, Pt. 2A, Ann. Rept. 1919, 45 p.
- 1927, Fensters in the Cumberland overthrust block in southwestern Virginia: *Va. Geol. Survey Bull.* 28, 12 p.
- Butts, Charles, and Nelson, W. A., 1925, Geology and mineral resources of the Crossville quadrangle: *Tenn. Geol. Survey Bull.* 33-D, 41 p.
- Campbell, M. R., 1899, Standingstone quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 53
- Glenn, L. C., 1925, The northern Tennessee coal field: *Tenn. Div. Geology Bull.* 33-B, 478 p.
- Hayes, C. W., 1894a, Kingston quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 4
- 1894b, Chattanooga quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 6
- 1894c, Sewanee quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 8
- 1895a, Pikeville quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 21
- 1895b, McMinnville quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 22
- Jillson, W. R., 1923, Emory River overthrust of east Tennessee: *Pan-American Geologist*, v. 39, p. 373-378
- Keith, Arthur, 1896, Briceville quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 33
- 1897, Wartburg quadrangle, Tennessee: *U. S. Geol. Survey Geol. Atlas*, Folio 40
- Miller, R. L., and Fuller, J. O., 1954, Geology and oil resources of the Rose Hill district—the Fenster area of the Cumberland Overthrust block—Lee County, Virginia: *Va. Geol. Survey Bull.* 71, 383 p.
- Nelson, W. A., 1925a, The southern Tennessee coal field: *Tenn. Div. Geology Bull.* 33-A, 239 p.
- 1925b, The coals and geology of the Herbert Domain: *Tenn. Div. Geology Bull.* 33-C, 54 p.
- Rich, J. L., 1934, Mechanics of low-angle overthrust faulting as illustrated by Cumberland thrust block, Virginia, Kentucky, and Tennessee: *Am. Assoc. Petroleum Geologists Bull.*, v. 18, p. 1584-1596
- Rodgers, John, 1950, Mechanics of Appalachian folding as illustrated by Sequatchie anticline, Tennessee and Alabama: *Am. Assoc. Petroleum Geologists Bull.*, v. 34, p. 672-681
- 1953a, Geologic map of East Tennessee with explanatory text: *Tenn. Div. Geology Bull.* 58, pt. II
- 1953b, The folds and faults of the Appalachian Valley and Ridge province, in McGrain, Preston, *Editor*, Southeastern mineral symposium 1950: *Ky. Geol. Survey*, ser. 9, Spec. Pub. 1, p. 150-166
- Safford, J. M., 1869, Geology of Tennessee: Nashville, S. C. Mercer, p. 137-143
- Stearns, R. G., 1954, The Cumberland Plateau overthrust and geology of the Crab Orchard Mountains area, Tennessee: *Tenn. Div. Geology Bull.* 60, 47 p.
- 1955, Low-angle overthrusting in central Cumberland Plateau, Tennessee: *Am. Assoc. Petroleum Geologists Bull.*, v. 66, p. 615-628
- Swingle, G. D., Hardeman, W. D., and Stearns, R. G., 1956, Structural relationship of the Valley and Ridge and Cumberland Plateau in the Southern Appalachians (Abstract): *Geol. Soc. America Bull.*, v. 67, no. 12, pt. 2, p. 1758
- Wentworth, C. K., 1921, Russell Fork fault of southwest Virginia: *Jour. Geology*, v. 29, p. 351-369
- Wilson, C. W., Jr., 1949, Pre-Chattanooga stratigraphy in Central Tennessee: *Tenn. Div. Geology Bull.* 56, 407 p.
- Wilson, C. W., Jr., Jewell, J. W., and Luther, E. T., 1956, Pennsylvanian geology of the Cumberland Plateau, Tennessee: *Tenn. Div. Geology Bull.* 58, 21 p.
- VANDERBILT UNIVERSITY, NASHVILLE, TENNESSEE;
TENNESSEE DIVISION OF GEOLOGY, NASHVILLE,
TENNESSEE
- MANUSCRIPT RECEIVED BY THE SECRETARY OF THE
SOCIETY, FEBRUARY 24, 1958